

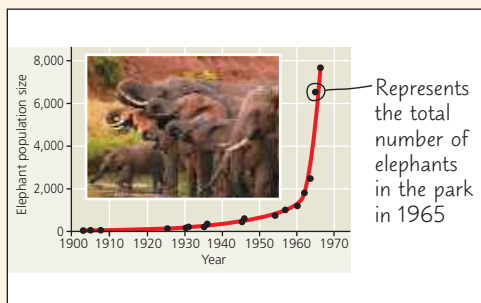
53 Population Ecology

KEY CONCEPTS

- 53.1** Biotic and abiotic factors affect population density, dispersion, and demographics *p. 1191*
- 53.2** The exponential model describes population growth in an idealized, unlimited environment *p. 1196*
- 53.3** The logistic model describes how a population grows more slowly as it nears its carrying capacity *p. 1197*
- 53.4** Life history traits are products of natural selection *p. 1200*
- 53.5** Density-dependent factors regulate population growth *p. 1202*
- 53.6** The human population is no longer growing exponentially but is still increasing extremely rapidly *p. 1207*

Study Tip

Annotate a graph: This chapter contains several *scatter plots*, graphs in which each measurement or observation is plotted as a single point. To make sure you understand a scatter plot, select one point and describe what it represents, as shown here for Figure 53.8. Do this for Figures 53.4, 53.8, 53.10, 53.16, and 53.18.



Go to Mastering Biology

For Students (in eText and Study Area)

- Get Ready for Chapter 53
- Table 53.1 Walkthrough: Life Table for Female Belding's Ground Squirrels
- BioFlix® Animation: Population Ecology

For Instructors to Assign (in Item Library)

- Interpreting Data: Survivorship Curves
- Interpreting Data: Human Population Growth



Figure 53.1 Although it was once common along the Atlantic coast, by 1985 prospects looked grim for the piping plover (*Charadrius melodus*). Since then, nesting site protection has enabled the piping plover's numbers to grow, with recent surveys finding 1,400 to 4,000 adults in recent years. Such variation is typical of natural populations, all of which vary in number (size) over time.

What affects the size of a population and how it changes over time?

Adding individuals to or **removing** individuals from a population **changes the population size:**

